SHANGHAI JIAO TONG UNIVERSITY

APEX 数据和知识管理实验室
DATA & KNOWLEDGE MANAGEMENT LAB

UCL

When Large Language Models Meet Agents

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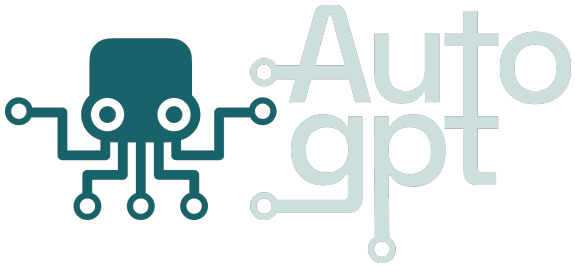
https://jinjiarui.github.io/

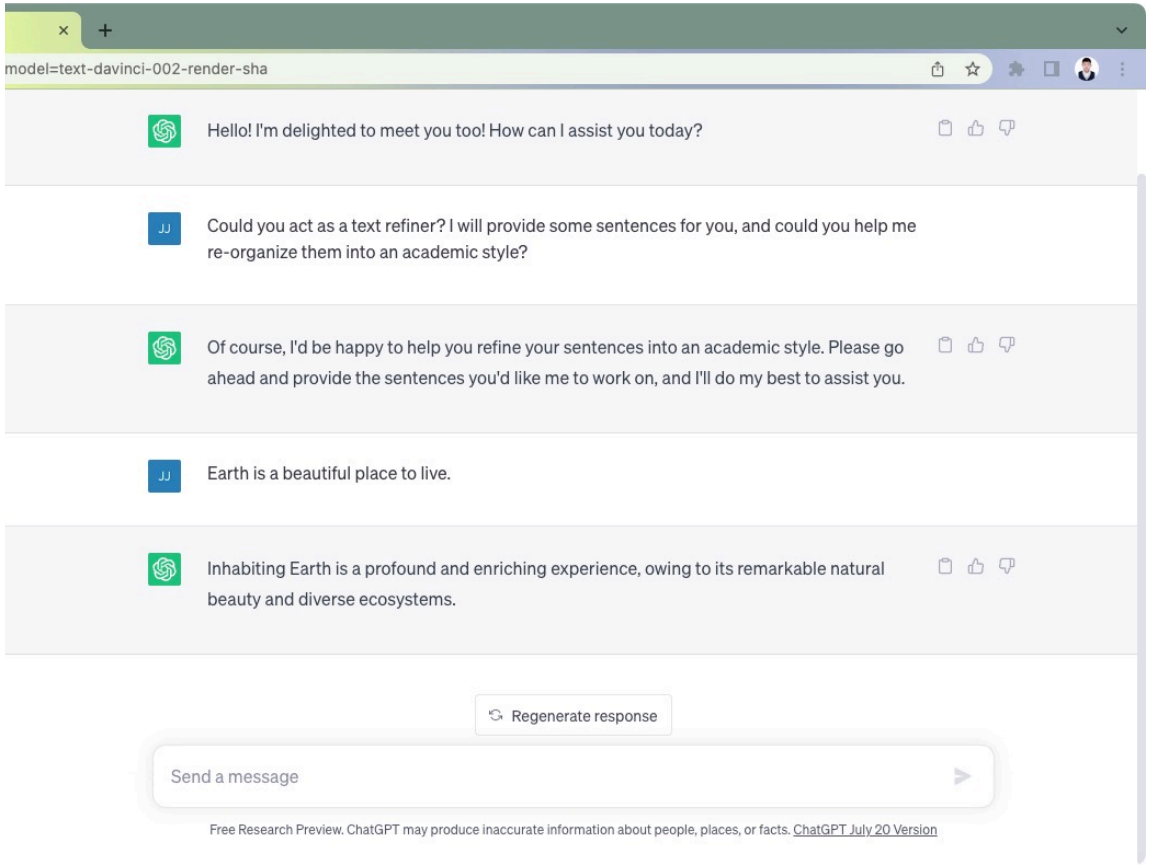
15:00-16:00, December 27, 2023
Meituan, Shanghai

Slides: https://jinjiarui.github.io/project/files/ lm4agent_meituan.pdf

PART I: Applications and Products

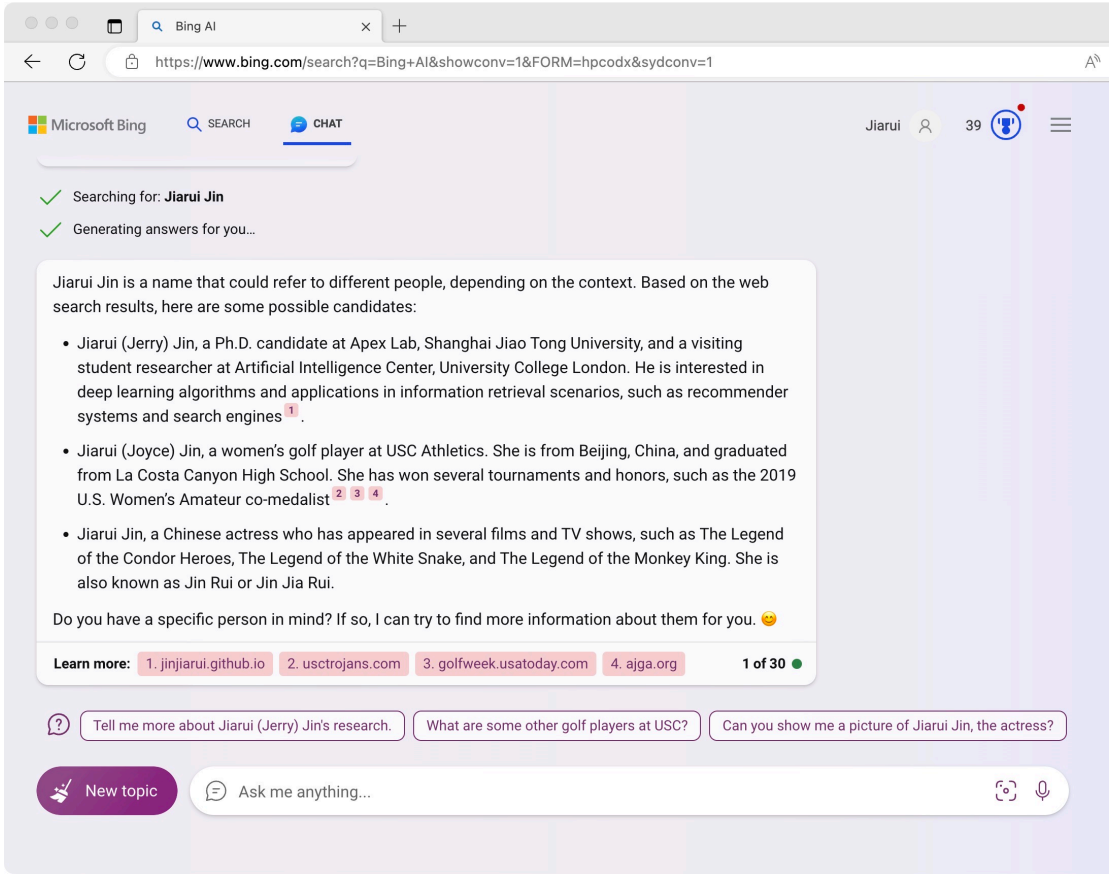


ChatGPT: Vanilla LLM (ChatGPT Plus: LLM + Tools)



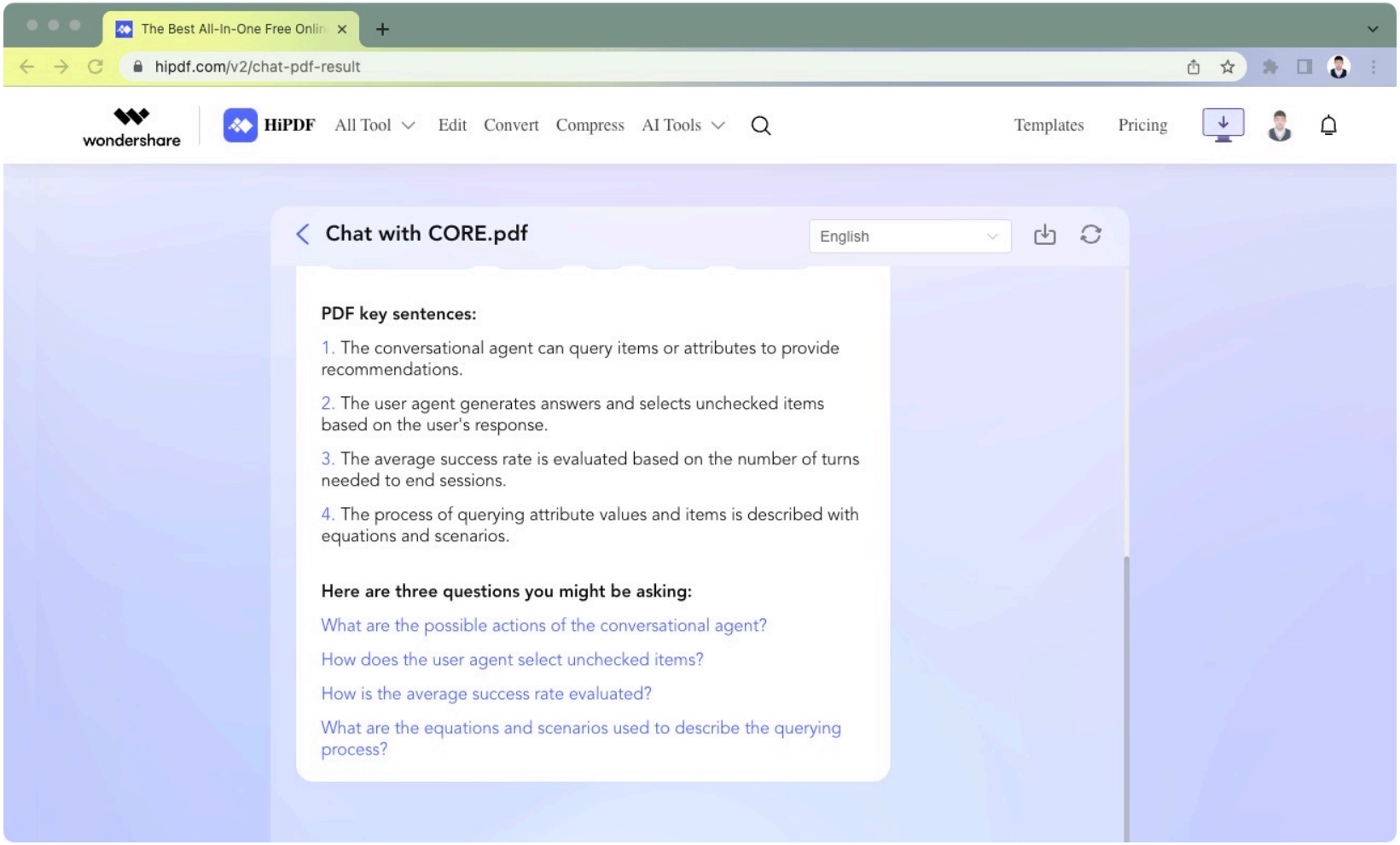
LLM can communicate like human and well do many language-related tasks such as information extractor or question answer.

New Bing: LLM + Memory (Web)
(New Bing v2: LLM + Memory + Tools)



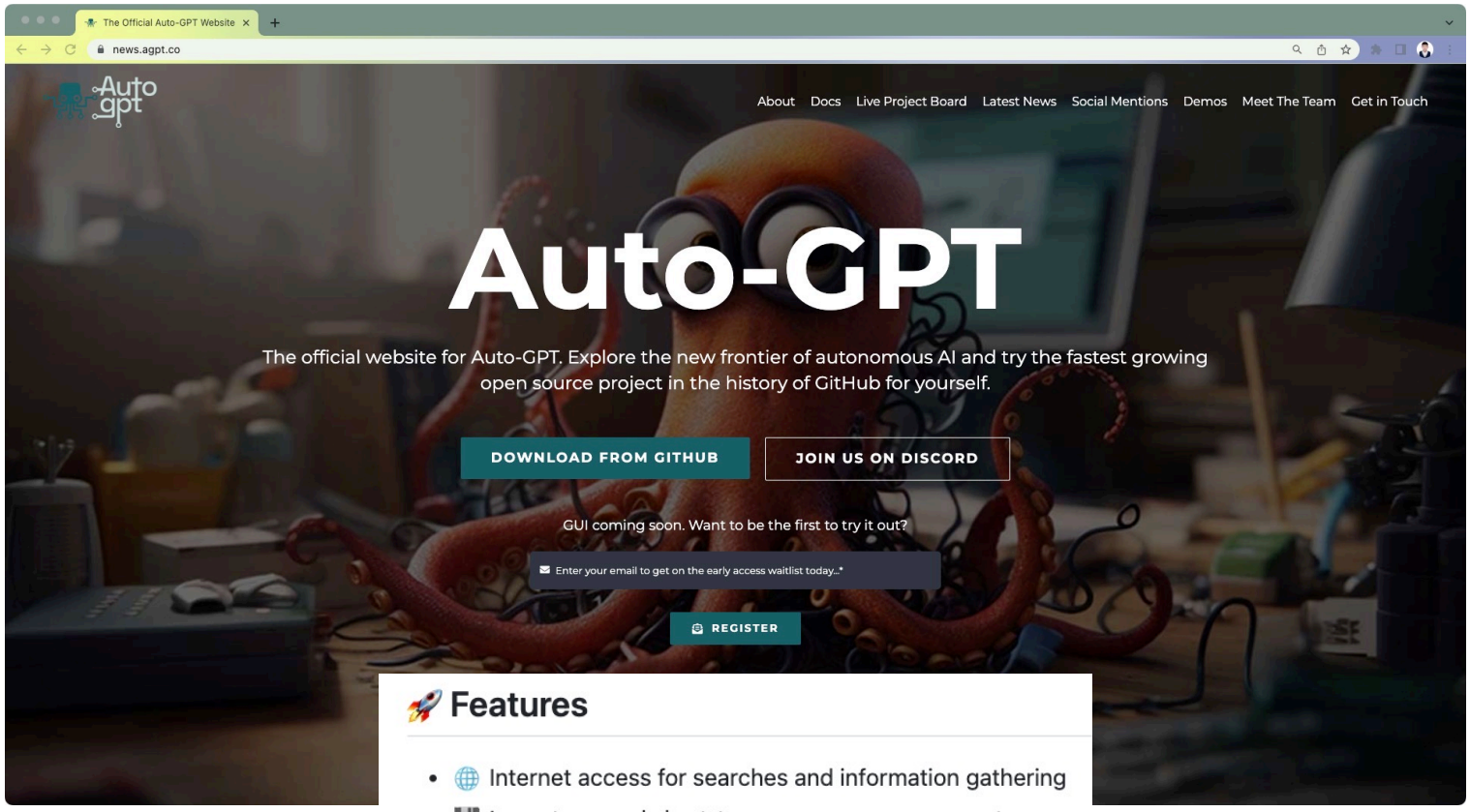
retrieve from database (or website) and organize in human-like language by LLM.

Chat PDF: LLM + Memory (PDF)

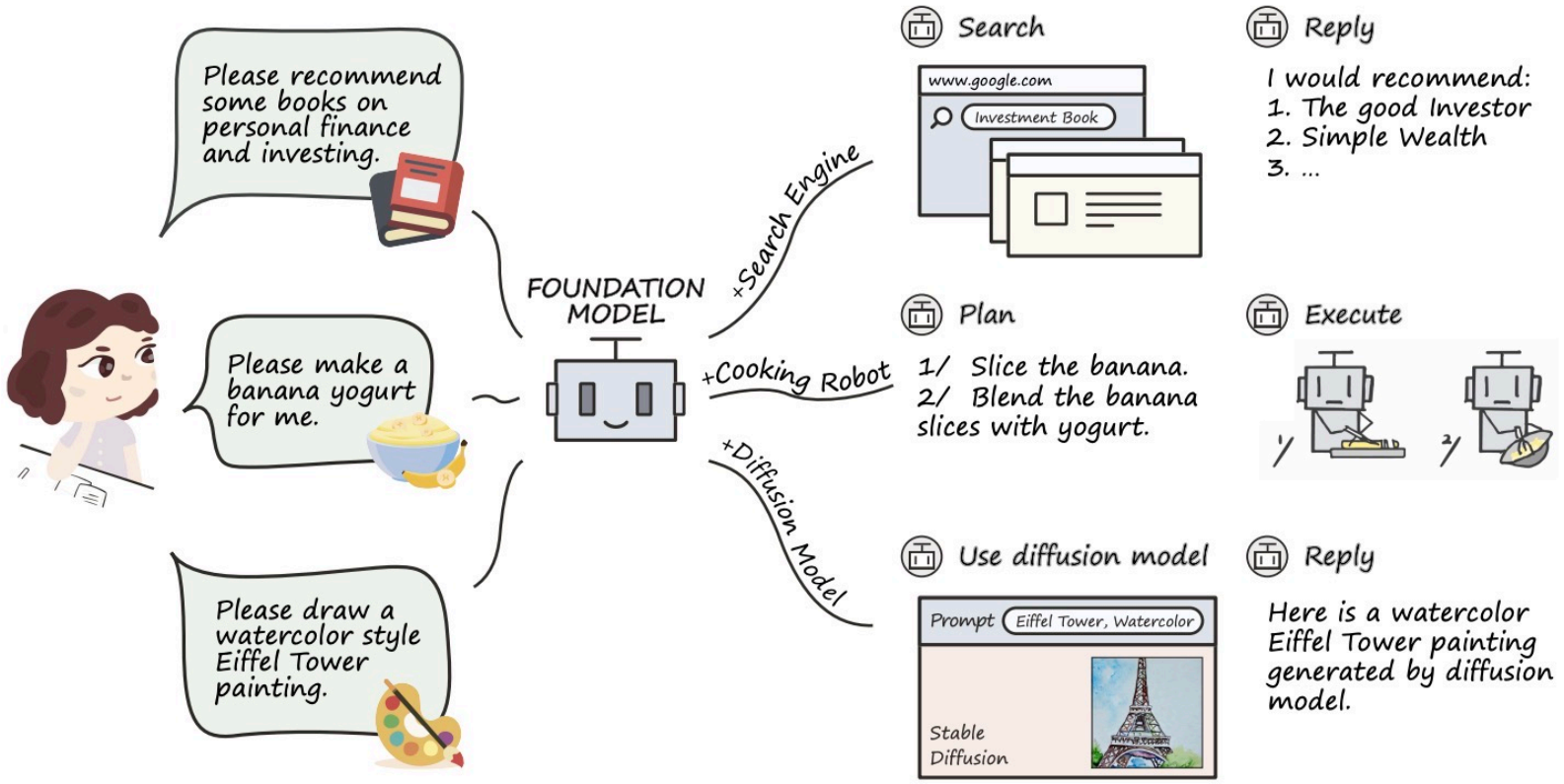


extract key information and summarize them into human language.

Vector Database: LLM + Memory



Agent = LLM + Memory + Tools + …



Outline

- PART I: Applications and Products
- **PART II: LLM + Memory (Extra Database)**
- **PART III: LLM + Tools (Extra Tools)**
- PART IV: Take-Aways

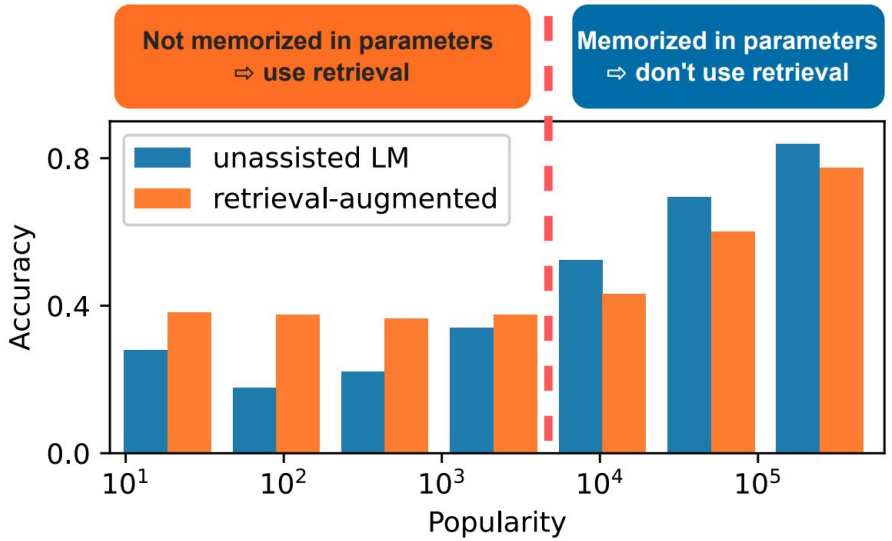
PART II: LLM + Memory

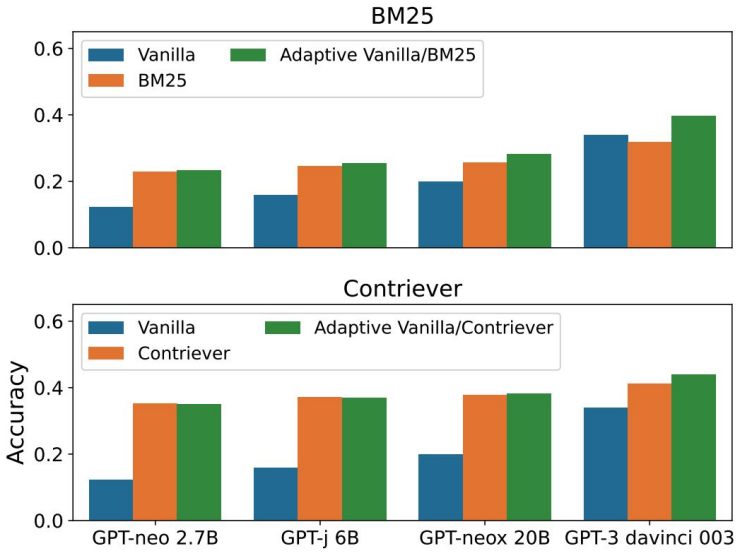
GUNDAM

Golden plUg-iN DAta Manager

Background: Some Observations

- Retrieved documents from external database or website can significantly improve the performance (i.e., accuracy and stability) of large language models, especially for questions whose popularity is lower than a threshold.

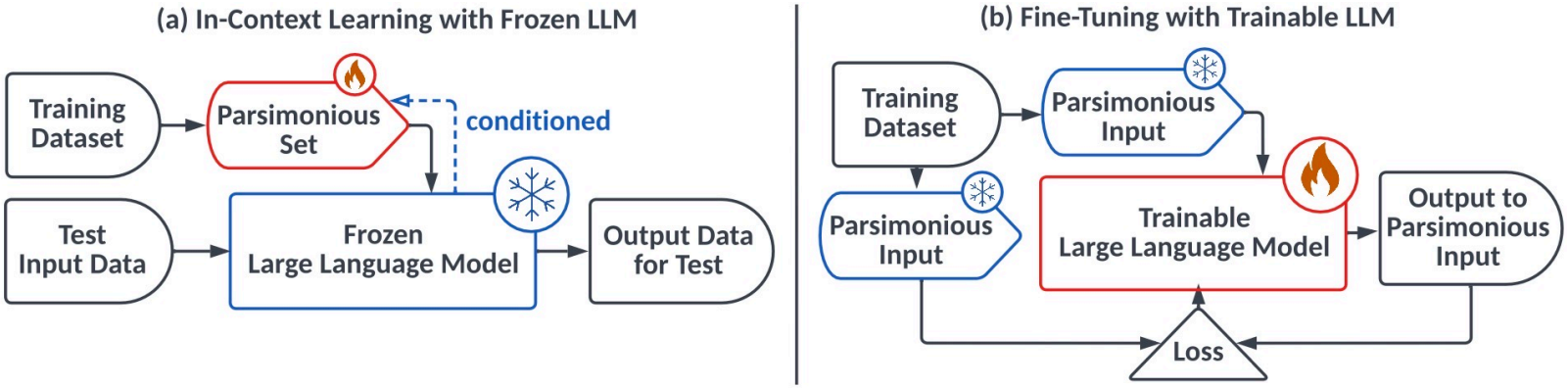




Ref: When Not to Trust Language Models: Investigating Effectiveness and Limitations of Parametric and Non-Parametric Memories. https://arxiv.org/pdf/2212.10511.pdf. arXiv.

Background: Some Observations

- In-context learning is using relevant documents as prompt (called demonstrations), which is an unsupervised learning paradigm.
- Fine-tuning always needs large corpus of textual data, but warm-up, another supervised learning paradigm, only pushes large language models go through the whole data corpus once.



Ref: Manage Your Plug-in Data for Language Models: A Data-Centric Approach. https://jinjiarui.github.io/preprints/GUNDAM.pdf. Pre-print.

Our Idea

- Idea is to synthesize a smaller, golden subset $\mathcal{D}^{\mathrm{g}}$ that can achieve comparable (or even better) performance of using the whole training dataset $\mathcal{D}$.
- Outer-loop: optimize $\mathcal{D}^{\mathrm{g}}$ with a frozen large language model φ

$$\mathrm{argmin}_{\mathcal{D}^{\mathrm{g}}} \mathbb{E}_{(q,a)\sim\mathcal{D}} [l(\varphi_{\mathcal{D}^{\mathrm{g}}}(q), a)]$$

- Inner-loop: optimize φ for a fixed $\mathcal{D}^{\mathrm{g}}$:

$$\mathrm{argmin}_{\varphi} \mathbb{E}_{(q,a)\sim\mathcal{D}^{\mathrm{g}}} [l(\varphi(q), a)]$$

- This idea is actually a data-centric perspective (data quality outweighs data quantity) of retrieval-augmented large language model, where $\mathcal{D}^{\mathrm{g}}$ is high-quality data.

Ref: Manage Your Plug-in Data for Language Models: A Data-Centric Approach. https://jinjiarui.github.io/preprints/GUNDAM.pdf. Pre-print.

What is High-Quality Data

- High-quality data should be both sufficient and necessary, where sufficiency and necessity are **conditioned on the given large language model**.

Definition 1 (**Sufficient Plug-in Instance**). *Given tuple* (X, Y, Z, S)*, we say that a training example* $(\boldsymbol{x}_n, \boldsymbol{y}_n)$ *is a sufficient plug-in instance for a target data point* $(\boldsymbol{x}_m, \boldsymbol{y}_m)$ *(namely plugging-in* $(\boldsymbol{x}_n, \boldsymbol{y}_n)$ *is sufficient to get the right answer of* $\boldsymbol{x}_m$*), if the following equation holds:*

$$Y_{\boldsymbol{x}_m} = 1 | \texttt{plug}((\boldsymbol{x}_n, \boldsymbol{y}_n)); Z = \emptyset, S = (Y_{\boldsymbol{x}_m} = 0). \tag{5}$$

It indicates that before plugging-in $(\boldsymbol{x}_n, \boldsymbol{y}_n)$*, there is no plugged-in data (i.e.,* $Z = \emptyset$*) and the LM's answer to* $\boldsymbol{x}_m$ *is wrong (i.e.,* $S = (Y_{\boldsymbol{x}_m} = 0)$*), and when we plug-in* $(\boldsymbol{x}_n, \boldsymbol{y}_n)$ *(i.e.,* $\texttt{plug}((\boldsymbol{x}_n, \boldsymbol{y}_n))$*), then it would correct the LM's answer to* $\boldsymbol{x}_m$ *(i.e.,* $Y_{\boldsymbol{x}_m} = 1$*).*

Definition 2 (**Necessary Plug-in Instance**). *Given tuple* (X, Y, Z, S)*, we say that a training example* $(\boldsymbol{x}_n, \boldsymbol{y}_n)$ *is a necessary plug-in instance for a target data point* $(\boldsymbol{x}_m, \boldsymbol{y}_m)$ *(namely plugging-in* $(\boldsymbol{x}_n, \boldsymbol{y}_n)$ *is necessary to get the right answer of* $\boldsymbol{x}_m$*), if the following equation holds:*

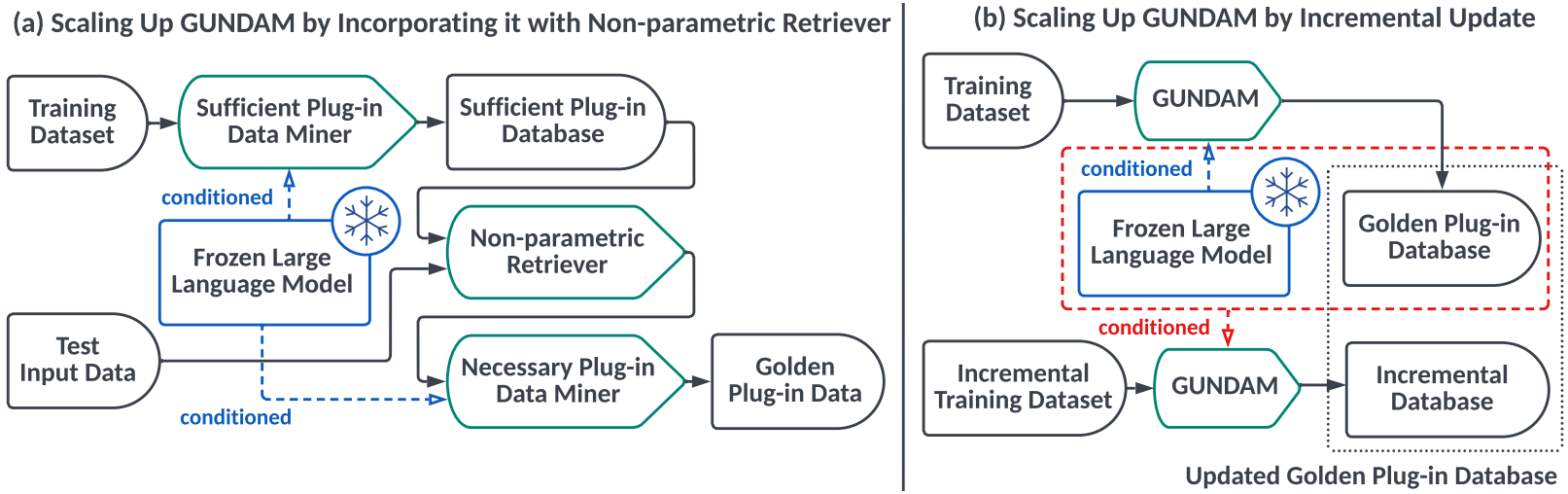
$$Y_{\boldsymbol{x}_m} = 0 | \texttt{unplug}((\boldsymbol{x}_n, \boldsymbol{y}_n)); Z = ((\boldsymbol{x}_n, \boldsymbol{y}_n)), S = (Y_{\boldsymbol{x}_m} = 1). \tag{6}$$

It means that before unplugging $(\boldsymbol{x}_n, \boldsymbol{y}_n)$*, there is plugged-in data* $(\boldsymbol{x}_n, \boldsymbol{y}_n)$ *(i.e.,* $Z = ((\boldsymbol{x}_n, \boldsymbol{y}_n))$*) and the LM's answer to* $\boldsymbol{x}_m$ *is right (i.e.,* $S = (Y_{\boldsymbol{x}_m} = 1)$*), and when we unplug* $(\boldsymbol{x}_n, \boldsymbol{y}_n)$ *(i.e.,* $\texttt{unplug}((\boldsymbol{x}_n, \boldsymbol{y}_n))$*), then it would cause the LM's answer to* $\boldsymbol{x}_m$ *being wrong (i.e.,* $Y_{\boldsymbol{x}_m} = 0$*).*

Ref: Manage Your Plug-in Data for Language Models: A Data-Centric Approach. https://jinjiarui.github.io/preprints/GUNDAM.pdf. Pre-print.

Scaling Up GUNDAM

- Adopt a non-parametric retriever (e.g., random, similarity-based, diversity-based retrievers) to meet the window length of large language models.
- Regard selected data and large language models as a whole to enable incremental update.



Ref: Manage Your Plug-in Data for Language Models: A Data-Centric Approach. https://jinjiarui.github.io/preprints/GUNDAM.pdf. Pre-print.

Some Unsupervised Learning Results

Table 1: Result comparisons on text classification datasets for different LM bases (MED, LAR, NEO) with different retrievers (RAN, SIM, DIV) in different shot settings (1, 2, 5, 10). All the results are calculated based on 5 different random seeds, and the bold numbers mean using $\mathcal{D}_{\texttt{GOLD}}$ as $\widetilde{\mathcal{D}}_{\texttt{TRAIN}}$ can obtain at least 2% improvements than using $\mathcal{D}_{\texttt{TRAIN}}$, or vice versa. See Table A1 for more extended results (on datasets SUBJ, and FPB).

$\Psi_{\texttt{LM}}(\cdot)$	$\widetilde{\mathcal{D}}_{\texttt{TRAIN}}$	n	SST-2			SST-5			COLA			TREC		
			RAN	SIM	DIV	RAN	SIM	DIV	RAN	SIM	DIV	RAN	SIM	DIV
MED	$\mathcal{D}_{\texttt{TRAIN}}$	1	48.9	24.5	24.5	14.5	22.7	22.7	29.0	**38.8**	**38.8**	19.4	42.8	42.8
		2	51.2	65.7	62.5	18.0	25.6	23.7	30.9	38.5	36.2	21.4	57.2	51.4
		5	62.6	79.4	61.7	26.5	32.3	27.8	**69.4**	49.3	47.0	37.6	66.0	61.4
		10	50.9	83.8	76.9	14.9	35.3	30.4	31.6	52.5	58.8	**53.0**	71.4	65.8
	$\mathcal{D}_{\texttt{GOLD}}$	1	49.8	**48.1**	**48.1**	15.4	23.7	23.7	29.4	35.1	35.1	**37.4**	**48.4**	**48.4**
		2	**67.3**	**67.7**	**64.7**	**20.9**	**27.9**	**25.8**	31.1	**41.7**	34.9	**27.6**	56.8	52.2
		5	**70.3**	77.9	**68.5**	**28.6**	33.2	27.4	65.2	**57.3**	**54.6**	**40.8**	67.4	61.8
		10	**75.2**	83.0	77.2	**17.6**	36.2	29.8	**69.3**	**68.7**	**68.5**	44.6	**74.6**	64.6
LAR	$\mathcal{D}_{\texttt{TRAIN}}$	1	49.0	42.3	42.3	14.2	25.2	25.2	42.1	**38.3**	**38.3**	21.0	53.2	53.2
		2	68.0	70.7	59.6	18.1	29.7	24.4	**41.1**	36.8	37.7	28.2	62.6	60.6
		5	49.1	80.6	67.5	25.6	34.1	30.8	66.2	53.8	48.5	35.4	63.4	64.6
		10	71.1	84.6	73.1	28.7	38.7	36.6	43.4	55.5	56.1	43.2	66.0	68.8
	$\mathcal{D}_{\texttt{GOLD}}$	1	49.1	**47.7**	**47.7**	**18.7**	25.5	25.5	**49.6**	35.1	35.1	17.4	52.6	52.6
		2	67.8	**73.0**	61.2	**25.2**	29.7	24.1	38.1	36.6	37.0	**34.6**	62.2	59.4
		5	**59.3**	80.9	**69.6**	**39.3**	35.2	31.0	**69.2**	**68.6**	**66.6**	**45.4**	**65.5**	63.7
		10	**76.0**	84.7	**75.6**	**39.6**	38.8	37.1	**69.3**	**68.8**	**68.9**	**55.8**	**70.4**	68.6
NEO	$\mathcal{D}_{\texttt{TRAIN}}$	1	49.2	33.8	33.8	12.8	20.2	20.2	25.5	36.5	36.5	11.0	57.2	57.2
		2	76.8	81.5	76.3	17.9	26.9	22.7	30.7	55.5	56.5	17.6	52.6	42.2
		5	65.1	80.8	66.1	19.0	29.2	25.1	40.0	55.9	52.5	25.2	66.4	61.8
		10	69.8	84.1	69.7	12.7	33.7	31.9	69.6	59.3	63.4	41.6	70.6	69.0
	$\mathcal{D}_{\texttt{GOLD}}$	1	49.3	**48.3**	**48.3**	**18.5**	20.6	20.6	**48.3**	34.8	34.8	**18.2**	56.4	56.4
		2	75.1	82.6	**78.5**	**19.7**	27.4	22.8	**69.3**	**64.7**	**64.7**	**27.8**	54.0	**44.5**
		5	**73.2**	**82.9**	**71.6**	19.2	30.2	26.4	**68.7**	**67.2**	**65.8**	**50.4**	66.0	61.6
		10	**72.4**	85.8	**71.8**	**15.4**	**37.0**	**34.5**	69.8	**68.8**	**68.9**	**45.2**	**72.8**	68.8

Ref: Manage Your Plug-in Data for Language Models: A Data-Centric Approach. https://jinjiarui.github.io/preprints/GUNDAM.pdf. Pre-print.

Some Supervised Learning Results

Table 2: Performance comparisons on text classification datasets for the fine-tuning setting. We report both the mean and variance of accuracy using four different seeds and four different permutations of n-shots. See Table 7 for more extended results on datasets COLA, SUBJ, and FPB.

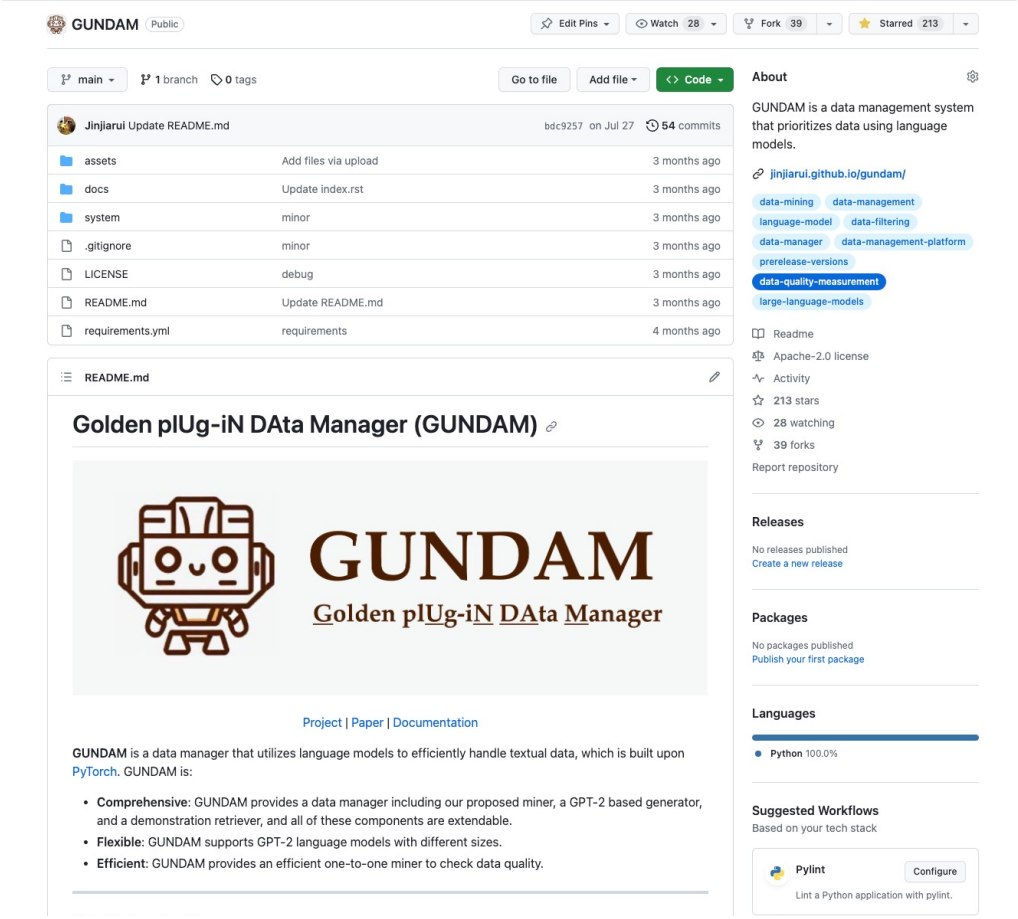
$\Psi_{\text{LLM}}(\cdot)$	$\widetilde{\mathcal{D}}_{\text{TRAIN}}$	n	SUBJ			SST-5			TREC		
			RAN	SIM	DIV	RAN	SIM	DIV	RAN	SIM	DIV
NEO	$\mathcal{D}_{\text{TRAIN}}$	1	72.7 (5.7)	91.0 (0.0)	91.0 (0.0)	17.3 (4.1)	24.6 (0.0)	24.6 (0.0)	63.3 (5.2)	79.5 (0.0)	79.5 (0.0)
		2	74.1 (4.6)	93.7 (0.0)	92.1 (0.0)	24.5 (3.2)	25.8 (0.0)	26.4 (0.0)	63.5 (5.7)	57.2 (0.0)	51.4 (0.0)
		5	70.8 (5.1)	93.3 (0.0)	92.7 (0.0)	23.6 (4.1)	27.8 (0.0)	27.3 (0.0)	67.8 (4.7)	66.6 (0.0)	73.0 (0.0)
		10	89.2 (4.4)	94.0 (0.0)	91.6 (0.0)	26.8 (2.9)	26.5 (0.0)	27.8 (0.0)	66.9 (3.8)	68.8 (0.0)	72.4 (0.0)
	$\mathcal{D}_{\text{GOLD}}$	1	**93.0** (4.3)	**93.5** (1.8)	**93.5** (1.8)	**19.5** (3.1)	**26.7** (2.0)	**26.7** (2.0)	**64.6** (3.2)	**80.6** (0.8)	**80.6** (0.8)
		2	**96.1** (3.8)	**94.1** (1.3)	**92.6** (1.2)	**25.6** (2.7)	**26.4** (0.7)	**28.6** (0.8)	**64.2** (3.7)	**79.6** (0.7)	**80.3** (0.9)
		5	**85.7** (3.5)	**94.7** (1.5)	**94.1** (1.1)	**27.4** (2.9)	**29.5** (1.8)	**29.7** (1.5)	**70.8** (3.2)	**78.2** (2.3)	**79.6** (1.9)
		10	**90.5** (3.3)	**95.5** (1.3)	**95.6** (1.4)	**28.9** (2.0)	**28.6** (1.7)	**29.0** (1.6)	**69.7** (2.7)	**71.2** (1.7)	**76.7** (1.9)

$\Psi_{\text{LLM}}(\cdot)$	$\widetilde{\mathcal{D}}_{\text{TRAIN}}$	n	FPB			SST-2			COLA		
			RAN	SIM	DIV	RAN	SIM	DIV	RAN	SIM	DIV
NEO	$\mathcal{D}_{\text{TRAIN}}$	1	62.7 (5.7)	78.4 (0.1)	78.4 (0.1)	41.3 (4.1)	52.6 (0.2)	52.8 (0.2)	49.3 (5.2)	68.5 (0.2)	68.5 (0.2)
		2	63.1 (4.6)	74.2 (0.3)	73.1 (0.2)	74.5 (3.2)	75.8 (0.4)	76.4 (0.5)	70.8 (5.7)	63.9 (0.2)	64.3 (0.4)
		5	70.8 (5.1)	73.3 (0.1)	72.7 (0.2)	53.6 (4.1)	57.8 (0.3)	57.3 (0.2)	30.7 (4.7)	54.4 (0.3)	54.0 (0.3)
		10	62.2 (4.4)	63.0 (0.6)	69.6 (0.5)	50.8 (2.9)	55.5 (0.2)	57.8 (0.2)	30.7 (3.8)	50.7 (0.3)	47.6 (0.4)
	$\mathcal{D}_{\text{FEED}}$	1	**73.0** (4.4)	**83.5** (1.5)	**83.5** (1.5)	**49.5** (4.1)	**56.7** (1.5)	**56.7** (1.5)	**56.8** (3.3)	**72.6** (0.9)	**72.6** (0.9)
		2	**76.1** (3.8)	**84.1** (1.4)	**82.5** (1.7)	**75.6** (2.8)	**76.4** (0.6)	**75.2** (0.5)	**74.2** (3.7)	**71.3** (0.7)	**71.5** (0.9)
		5	**75.7** (3.5)	**78.7** (1.5)	**83.1** (1.9)	**67.4** (2.9)	**69.5** (1.8)	**68.7** (1.9)	**71.7** (3.2)	**69.4** (2.3)	**70.0** (1.9)
		10	**70.5** (3.3)	**75.6** (1.3)	**77.6** (1.8)	**68.9** (2.0)	**68.6** (1.6)	**69.0** (1.4)	**71.3** (2.7)	**68.5** (1.7)	**68.5** (1.9)

Ref: Manage Your Plug-in Data for Language Models: A Data-Centric Approach. https://jinjiarui.github.io/preprints/GUNDAM.pdf. Pre-print.

Open-Source System

- Now we support GPT-2 and GPT-3 (medium, large, neo 1.3B) bases, and Llama 2 (7B).



Case Study

We consider the factual error made by Google Bard in the first demo[2]. Bard said *The James Webb Space Telescope took the very first pictures of a planet outside of our own solar system.* However, the fact is *The Very Large Telescope took the very first pictures of a planet outside of our own solar system.* Based on the above statements, we produce the question *What took the very first pictures of a planet outside of our own solar system?*

Then, we ask the question to `gpt-3.5-turbo`. The corresponding answer is:

```
The first pictures of a planet outside of our own solar system were taken by
the Hubble Space Telescope in 2008. The planet, named Fomalhaut b, orbits the
star Fomalhaut, which is located about 25 light-years away from Earth in the
constellation Piscis Austrinus.
```

We can see that the above answer is wrong. We then prompt the fact into the LM, then the answer is:

```
The Very Large Telescope took the very first pictures of a planet outside of
our own solar system.
```

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Case Study

The above answer is right, which verifies our Assumption A1. We ask `gpt-3.5-turbo` to generate some sufficient statements for the fact. The following is its output:

```
1. The Very Large Telescope made history by capturing the first-ever images of
a planet beyond our solar system.
2. Thanks to the advanced technology of the Very Large Telescope, scientists
were able to capture the first-ever photographs of an exoplanet.
3. The groundbreaking images captured by the Very Large Telescope marked a
major milestone in the study of exoplanets.
4. The discovery of a planet outside of our solar system was made possible by
the cutting-edge capabilities of the Very Large Telescope.
5. The Very Large Telescope's ability to capture images of distant celestial
bodies allowed scientists to observe an exoplanet for the first time in history.
```

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Case Study

We also evaluate separately using the above statements as a prompt to `gpt-3.5-turbo`, and either one of them can lead to a right answer. We provide detailed answers as follows.

```
1. The Very Large Telescope took the very first pictures of a planet outside
of our own solar system.
2. The Very Large Telescope took the very first pictures of a planet outside
of our own solar system.
3. The Very Large Telescope took the very first pictures of a planet outside
of our own solar system.
4. The Very Large Telescope took the very first pictures of a planet outside
of our own solar system.
5. The Very Large Telescope took the very first pictures of a planet outside
of our own solar system.
```

We then prompt the above five statements into `gpt-j-6b`, then the answer is:

```
1. The first-ever images of a planet beyond our solar system have been captured
by the Very Large Telescope in Chile.
2. The Hubble Space Telescope.
3. A team of astronomers led by the University of Arizona's Michael Liu.
4. The Hubble Space Telescope.
5. The Very Large Telescope.
```

Ref: Manage Your Plug-in Data for Language Models: A Data-Centric Approach.
https://jinjiarui.github.io/preprints/GUNDAM.pdf. Pre-print.

PART III: LLM + Tools

CORE

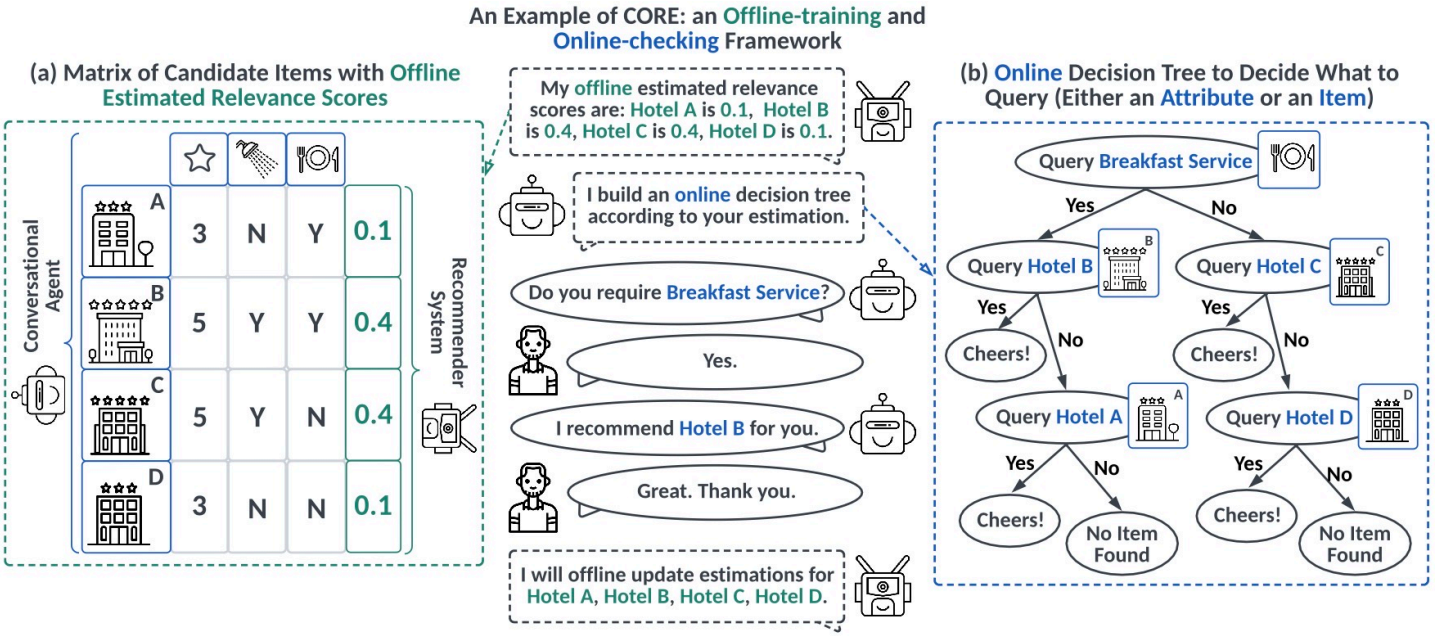
A Plug-and-Play COnversational Agent for REcommender System

Background

- Conversational recommender systems are designed to online query user preference on attributes and items.
- Existing solutions are bridging attribute recommendation and item recommendation components and conversational component into a whole system through a reinforcement learning framework.
- Drawback 1: People may not know her preference on an attribute, but she may know whether she likes a specific attribute value.
- Drawback 2: Many powerful large language models can be only access through APIs, can not be end-to-end optimized.

Our Idea

- Bridge recommender systems and conversational agents through an offline-training and online-checking framework, only requiring estimated scores from recommender systems and APIs from conversational agent.



Ref: Lending Interaction Wings to Recommender Systems with Plug-and-Play Conversational Agents. https://jinjiarui.github.io/preprints/CORE.pdf. NeurIPS 2023.

Online Decision Tree

- Metric: how many **relevance scores** have been checked by querying an item or an attribute.

Definition 1 (**Uncertainty and Certainty Gain**). *For the k-th turn, we define uncertainty, denoted as $\mathtt{U}_k$, to measure how many estimated relevance scores are still unchecked, i.e.,*

$$\mathtt{U}_k := \mathtt{SUM}(\{\Psi_{\mathtt{RE}}(v_m) | v_m \in \mathcal{V}_k\}), \tag{1}$$

where $\Psi_{\mathtt{RE}}(v_m)$[1] outputs the estimated relevance score for item v_m, $\mathcal{V}_k$ is the set of all the unchecked items after k interactions, and $\mathcal{V}_k$ is initialized as $\mathcal{V}_0 = \mathcal{V}$. Then, the certainty gain of k-th interaction is defined as $\Delta\mathtt{U}_k := \mathtt{U}_{k-1} - \mathtt{U}_k$, i.e., how many relevance scores are checked at the k-th turn. Since our goal is to find a target item, if $\Psi_{\mathtt{CO}}(\cdot)$ successfully finds one at the k-th turn, then we set all the items checked, namely $\mathtt{U}_k = 0$.

In this regard, our conversational agent is to minimize $\mathtt{U}_k$ at each k-th turn via removing those checked items from $\mathcal{V}_k$. Considering that online updating $\Psi_{\mathtt{RE}}(\cdot)$ is infeasible in practice due to the high latency and computation costs, the objective of $\Psi_{\mathtt{CO}}(\cdot)$ can be expressed as:

$$\min_{\Psi^*_{\mathtt{RE}}} K, \text{ s.t., } \mathtt{U}_K = 0, \tag{2}$$

where K is the number of turns, and $\Psi^*_{\mathtt{RE}}$ means that the recommender system is frozen. To this end, the design of our conversational agent could be organized as an uncertainty minimization problem.

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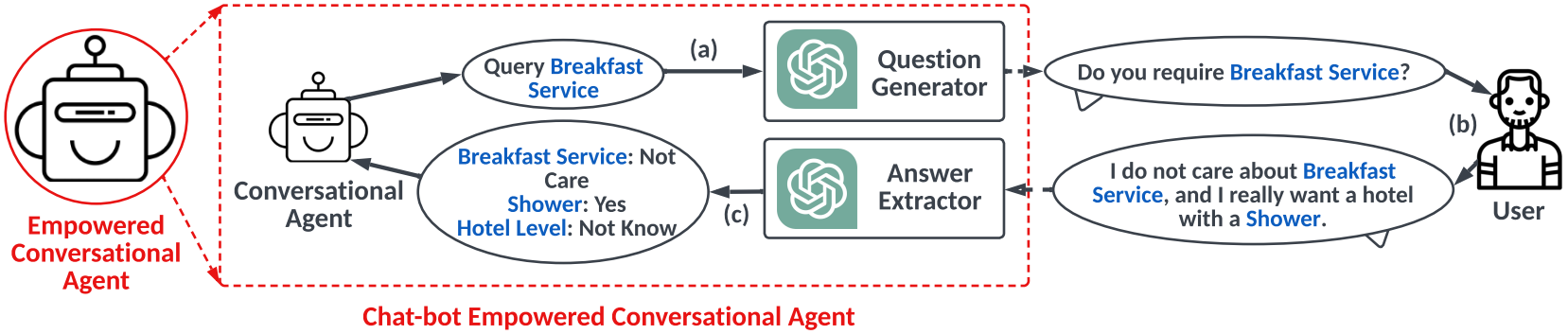
Prompt Design for Extractor and Generator



LLM can encode some latent features from conversations.

Ref: Lending Interaction Wings to Recommender Systems with Plug-and-Play Conversational Agents. https://jinjiarui.github.io/preprints/CORE.pdf. NeurIPS 2023.

Prompt Design for Extractor and Generator

```
"""
prompt = f"""
You will be provided with text delimited by triple quotes.
If you can infer the user preference on attributes,
then re-write the text in the following format:
[attribute name]: [user perference]
Attribute names include Breakfast Service, Hotel Level, and Shower.
User preference includes Yes to denote the positive preference, No to denote the
negative preference, and Not Care to denote the user does not care.
If you can not infer the user preference on attributes,
then re-write the text in the following format:
[attribute name]: Not Know
‘‘‘{text}‘‘‘
"""
response = get_completion(prompt)
print(response)
```

The following is the corresponding output by the LLM.

```
Breakfast Service: Not Care
Hotel Level: Not Know
Shower: Yes
```

Ref: Lending Interaction Wings to Recommender Systems with Plug-and-Play Conversational Agents. https://jinjiarui.github.io/preprints/CORE.pdf. NeurIPS 2023.

Prompt Design for Extractor and Generator

```
# Large language model as question generator.
text = f"""
Attribute, Hotel Level is 5, Hotel
"""
prompt = f"""
You will be provided with text delimited by triple quotes.
If it starts with the word "item", it denotes an item \
then you should generate a text to recommend the item to the user.
Otherwise, it denotes an attribute \
then you should generate a text to query the user's preference on the attribute.
You should be gentle.
‘‘‘{text}‘‘‘
"""
response = get_completion(prompt)
print(response)
```

The following is the corresponding output by the LLM.

```
Excuse me, may I ask for your preference on hotel level? Would you prefer a
5-star hotel or are you open to other options?
```

Ref: Lending Interaction Wings to Recommender Systems with Plug-and-Play Conversational Agents. https://jinjiarui.github.io/preprints/CORE.pdf. NeurIPS 2023.

Prompt Design for Extractor and Generator

```
# Large language model as latent feature extractor.
text = f"""
I am a student, I do not care about breakfast service, and I really want a hotel with a shower.
"""
prompt = f"""
You will be provided with text delimited by triple quotes.
If you can infer the user preference on attributes,
then re-write the text in the following format:
[attribute name]: [user perference]
Attribute names include Breakfast Service, Hotel Level, Price, and Shower.
User preference includes Yes to denote the positive preference, No to denote the
negative preference, and Not Care to denote the user does not care.
If you can infer the user has a low budget, please make Price: Low.
If you can not infer the user preference on attributes,
then re-write the text in the following format:
[attribute name]: Not Know
‘‘‘{text}‘‘‘
"""
response = get_completion(prompt)
print(response)
```

The following is the corresponding output by the LLM.

```
Breakfast Service: Not Care
Hotel Level: Not Know
Price: Low
Shower: Yes
```

Ref: Lending Interaction Wings to Recommender Systems with Plug-and-Play Conversational Agents. https://jinjiarui.github.io/preprints/CORE.pdf. NeurIPS 2023.

Some Results

Table 2: Results comparison of querying attribute values on tabular datasets. See Table A2 for the full version.

$\Psi_{\texttt{RE}}(\cdot)$	$\Psi_{\texttt{CO}}(\cdot)$	Amazon				LastFM				Yelp			
		T@3	S@3	T@5	S@5	T@3	S@3	T@5	S@5	T@3	S@3	T@5	S@5
COLD START	AG	6.47	0.12	7.83	0.23	6.77	0.05	8.32	0.14	6.65	0.08	8.29	0.13
	ME	6.50	0.12	8.34	0.16	6.84	0.04	8.56	0.11	6.40	0.15	8.18	0.20
	CORE	**6.02**	**0.25**	**6.18**	**0.65**	**5.84**	**0.29**	**5.72**	**0.74**	**5.25**	**0.19**	**6.23**	**0.65**
	$\texttt{CORE}_{\texttt{D}}^{+}$	6.00	0.26	6.01	0.67	5.79	0.30	5.70	0.75	5.02	0.21	6.12	0.68
FM	AG	**2.76**	0.74	**2.97**	0.83	4.14	0.52	4.67	0.64	3.29	0.70	3.39	0.81
	CRM	4.58	0.28	6.42	0.38	4.23	0.34	5.87	0.63	4.12	0.25	6.01	0.69
	EAR	4.13	0.32	6.32	0.42	4.02	0.38	5.45	0.67	4.10	0.28	5.95	0.72
	CORE	3.26	**0.83**	3.19	**0.99**	**3.79**	**0.72**	**3.50**	**0.99**	**3.14**	**0.84**	**3.20**	**0.99**
	$\texttt{CORE}_{\texttt{D}}^{+}$	3.16	0.85	3.22	1.00	3.75	0.74	3.53	1.00	3.10	0.85	3.23	1.00
DEEP FM	AG	**3.07**	0.71	3.27	0.82	3.50	0.68	3.84	0.79	3.09	0.74	3.11	0.88
	CRM	4.51	0.29	6.32	0.40	4.18	0.38	5.88	0.63	4.11	0.23	6.02	0.71
	EAR	4.47	0.30	6.35	0.43	4.01	0.37	5.43	0.69	4.01	0.32	5.74	0.75
	CORE	3.23	**0.85**	**3.22**	**0.99**	**3.47**	**0.81**	**3.34**	**1.00**	**2.98**	**0.93**	**3.11**	**1.00**
PNN	AG	3.02	0.74	3.10	0.87	3.44	0.67	3.53	0.84	2.83	0.77	2.82	0.91
	CORE	**3.01**	**0.88**	**3.04**	**0.99**	**3.10**	**0.87**	**3.20**	**0.99**	**2.75**	**0.88**	**2.76**	**1.00**

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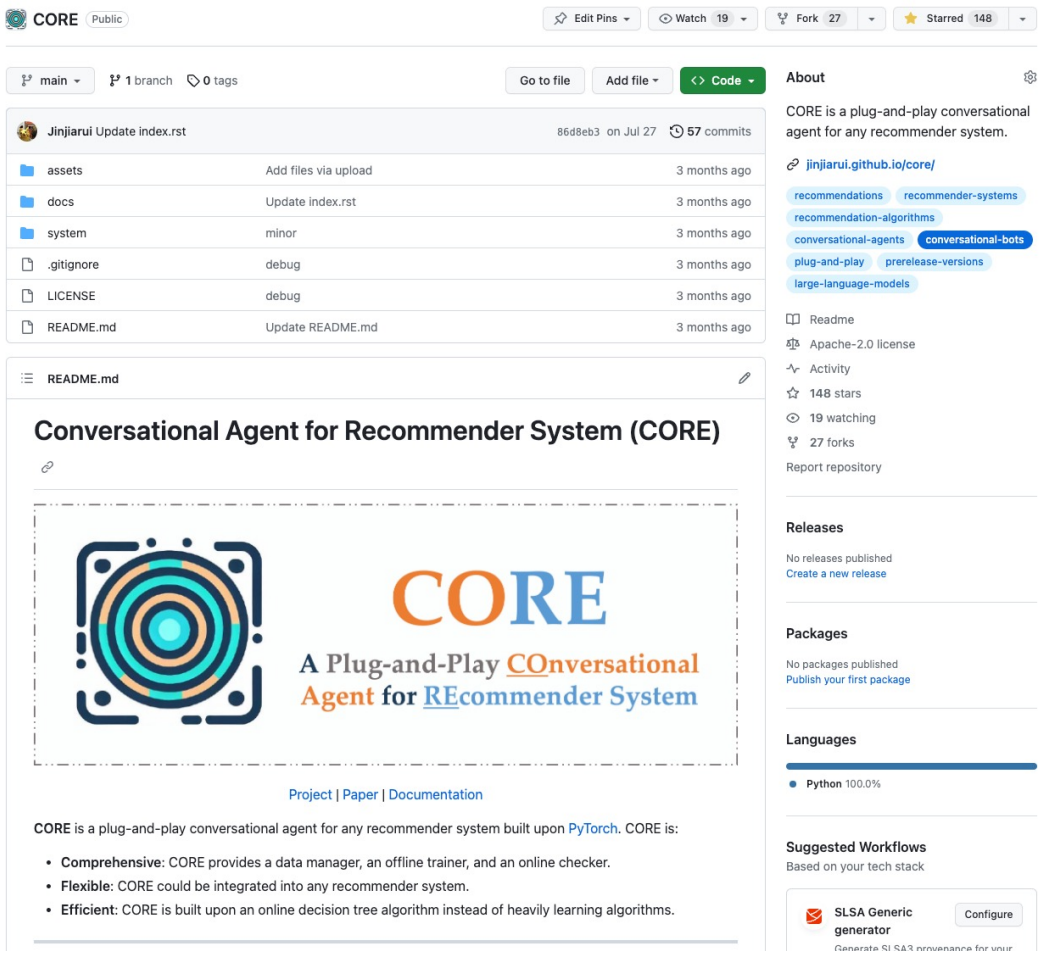
Open-Source System

- Now we support top-N recommendations for items and attribute values.



PART IV: Take-Aways

Summary and Future Directions

- Let study and use large language models as pets.
- Vanilla LLM can not well address some complex logical problems.
- GUDAM tells us that involving high-quality data can improve the LLM performance.
- CORE tells us that for each specific task, we can plug-in some APIs to help with LLM.

Prompt: How many times has Real Madrid won the Champions League?

Response: Real Madrid has won the UEFA Champions League a total of 13 times.

Prompt: Observation From Wikipedia
Title: 2021–22 UEFA Champions League
Content: Real Madrid defeated Liverpool 1–0 in the final, which was played at the Stade de France in Saint-Denis, France, for a record-extending 14th title, and their fifth in nine years .....

Response: I apologize for the error in my previous response. The information I provided was based on my knowledge cutoff date of September 2021. However, I have since been updated and can confirm that Real Madrid has now won the UEFA Champions League a total of 14 times.

Summary and Future Directions

- Let study and use large language models as pets.
- Vanilla LLM can not well address some complex logical problems.
- GUDAM tells us that involving high-quality data can improve the LLM performance.
- CORE tells us that for each specific task, we can plug-in some APIs to help with LLM.

Zero-shot Prompting: Here we provide a tool (API) "forecast_weather(city:str, N:int)", which could forecast the weather about a city on a specific date (after N days from today). The returned information covers "temperature", "wind", and "precipitation".
Please write codes using this tool to answer the following question: "What's the average temperature in Beijing next week?"

Few-shot Prompting: We provide some examples for using a tool. Here is a tool for you to answer question:
Question: "What's the temperature in Shanghai tomorrow?"

```
return forecast_weather("Shanghai", 1)["temperature"]
```

Question: "Will it rain in London in next two days?"

```
for i in range(2):
    if forecast_weather("London", i+1)["precipitation"] > 0:
        return True
return False
```

Question: "What's the average temperature in San Francisco next week?"

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Thanks for Your Listening

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Slides: https://jinjiarui.github.io/project/files/lm4agent_meituan.pdf